

P0004E Exercise 3 2025

Final answers

1. -

2. See exam solution P0004E 2024-08-28

3a $T_{PD} = \frac{6}{s^2 + 2s + 12}$

b $T_{PD} = \frac{58 + 10.8s}{s^2 + 12.8s + 64}$

c PD controller improves damping

d Use MATLAB 'step' command

4. a

b. $K_p = 3$

$K_I = 7.5 \times 10^{-3} \quad (T_i = 400)$

$K_D = 300 \quad (T_D = 100)$

5 a $K_{pc} = 5$

b. $K_p = 2.36 \left(\frac{50}{22} \right)$

$K_i = 1.36 \times 10^{-2} \quad (T_i = \frac{500}{3})$

6. Use MATLAB to check your answer

$$G(s) = \frac{20(s+1)}{s(s+5)(s^2+2s+10)}$$

$$= \frac{0.4(s+1)}{s\left(\frac{s}{5}+1\right)\left(\left(\frac{s}{\sqrt{10}}\right)^2 + \frac{2}{10}s+1\right)}$$

b. $PM = 104^\circ$ at 0.443 rad/s

$GM = 9.93$ at 4.01 rad/s

7. a Use MATLAB to draw the Bode plots

b. Observe that the Bode plot for the two T_Fs are almost identical or very similar for low frequency